

torch.bucketize#

<https://pytorch.org/docs/stable/generated/torch.bucketize.html#torch.bucketize>

Description:

Returns the indices of the buckets to which each value in the input belongs, where the boundaries of the buckets are set by boundaries. Return a new tensor with the same size as input. If right is False (default), then the left boundary is open. Note that this behavior is opposite the behavior of `numpy.digitize`. More formally, the returned index satisfies the following rules: returned index satisfies $\text{boundaries}[i-1] < \text{input}[m][n] \dots [l][x] \leq \text{boundaries}[i]$

Function Signature

```
torch.bucketize(input, boundaries, *, out_int32=False,
right=False, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

`out_int32` (bool, optional) – indicate the output data type. `torch.int32` if True, `torch.int64` otherwise. Default value is False, i.e. default output data type is `torch.int64`. `right` (bool, optional) – determines the behavior for values in boundaries. See the table above. `out` (Tensor, optional) – the output tensor, must be the same size as input if provided.

Code Examples

```
>>> boundaries = torch.tensor([1, 3, 5, 7, 9])
>>> boundaries
tensor([1, 3, 5, 7, 9])
>>> v = torch.tensor([[3, 6, 9], [3, 6, 9]])
>>> v
tensor([[3, 6, 9],
        [3, 6, 9]])
>>> torch.bucketize(v, boundaries)
tensor([[1, 3, 4],
        [1, 3, 4]])
>>> torch.bucketize(v, boundaries, right=True)
tensor([[2, 3, 5],
        [2, 3, 5]])
```

Detailed Documentation

Documentation

Returns the indices of the buckets to which each value in the input belongs, where the boundaries of the buckets are set by boundaries. Return a new tensor with the same size as input. If right is False (default), then the left boundary is open. Note that this behavior is opposite the behavior of `numpy.digitize`. More formally, the returned index satisfies the following rules:

returned index satisfies

$$\text{boundaries}[i-1] < \text{input}[m][n] \dots [l][x] \leq \text{boundaries}[i]$$
$$\text{boundaries}[i-1] \leq \text{input}[m][n] \dots [l][x] < \text{boundaries}[i]$$

input (Tensor or Scalar) – N-D tensor or a Scalar containing the search value(s).

boundaries (Tensor) – 1-D tensor, must contain a strictly increasing sequence, or the return value is undefined.

out_int32 (bool, optional) – indicate the output data type. `torch.int32` if True, `torch.int64` otherwise. Default value is False, i.e. default output data type is `torch.int64`.

right (bool, optional) – determines the behavior for values in boundaries. See the table above.

out (Tensor, optional) – the output tensor, must be the same size as input if provided.

